Microcontroller-Based Elevator Emulator

Brief

An elevator emulator implemented in C on an ATmega324A microcontroller, displayed on an 8×16 LED matrix showing four floors and a moving elevator car. Travellers appear on floors with colour-coded destination indicators, and the elevator navigates to pick them up and deliver them to the correct floor in sequence. Features include button and terminal-based floor selection, colour-coded traveller placement and movement with auto calculated routing, a seven-segment display showing current floor and direction of travel, adjustable elevator speed via a hardware switch, a piezo buzzer with distinct tones for traveller pickup and drop-off events. An IOBoard LED door animation sequence triggered on startup and arrival to floors. Terminal output provided live status updates including current floor, direction of travel, and floor counters for travel, both with and without passengers.

Demo Video

https://youtube.com/shorts/Tj7HGrwaXjg?feature=share